

Smartbright: Learning a Physics-Based Model of Light-Scattering With Symbolic Regression

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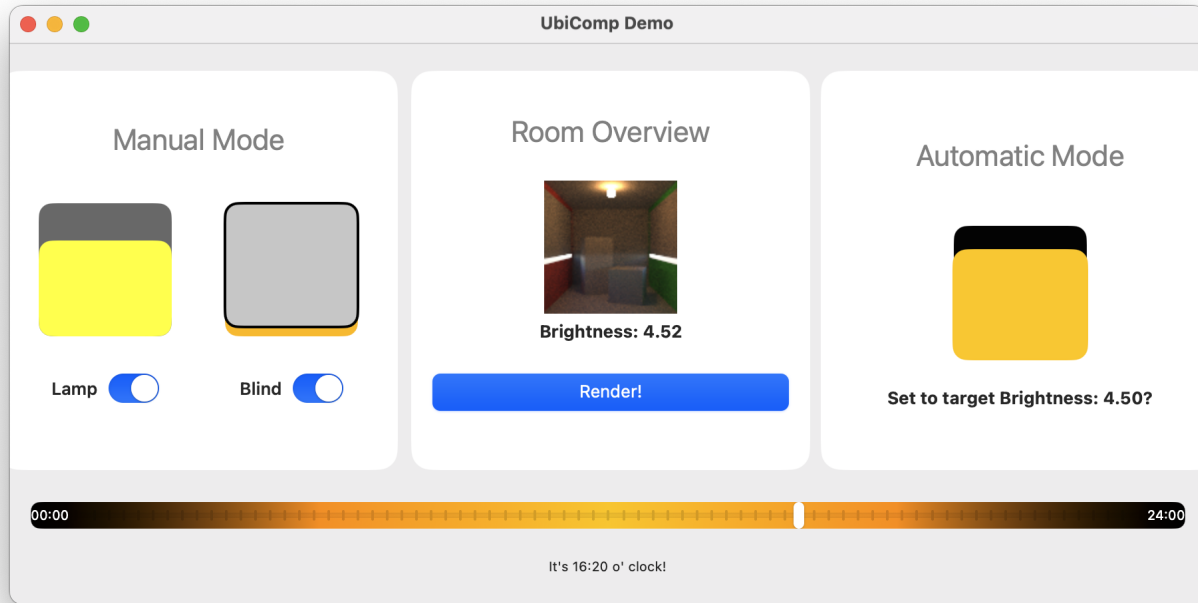


Figure 1: Manual smart home control on the left, compared to outcome-oriented smart home control as an implementation of our previous efforts on the right. Users can either set blinds and lamps in the smart home manually or let the smart home determine settings that achieve the specified brightness goal.

ABSTRACT

A smart home control application allows users to interact with the Cyber-Physical System (CPS) in their living space. Conventional applications offer manual control over individual smart devices, disregarding compound effects from using multiple devices. Manual control is not user-friendly, as users must explore smart device

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settings until achieving their target states. In previous works, we proposed a symbolic regression approach to find settings of smart devices that achieve user-specified goals. We now demonstrate a novel application, *SmartBright*, an outcome-oriented light control for macOS. *SmartBright* allows users to set a target brightness and the current time of day; the application finds settings for the window blind and lamp to achieve this goal. We evaluated *SmartBright* by comparing suggested settings for a bright room against ray-traced brightness throughout the day. Additionally, we applied the user preference *Do not use the lamp, if possible* to minimize lamp use and evaluated these settings. Results show *SmartBright* suggests settings achieving user goals throughout the day, with average errors of $0.1175 P_l$ for the unrestricted case, and $0.1730 P_l$ for the restricted case.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools; • **Computing methodologies** → Equation and inequality solving algorithms; • **Computer systems organization** → Embedded and cyber-physical systems.

KEYWORDS

Smart Home, Cyber-Physical Systems, Symbolic Regression, Ubiquitous Computing

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1 INTRODUCTION

Ubiquitous Cyber-Physical Systems (CPS), such as smart homes are increasingly prevalent. The primary advantage of a smart home lies in its CPS interface, which provides centralized control over various appliances, allowing users to modify environmental properties, like the brightness in a room. Mainstream interfaces for smart homes include voice-enabled applications and conventional apps on phones, tablets, or computers. However, these interfaces lack the capability to understand their environment and the effects of smart devices, especially those influencing multiple properties, such as AC units that alter both noise and temperature. Thus, users must manually adjust multiple devices to achieve desired target states.

In previous works we found that symbolic regression can be used to learn a physics-based model of a CPS, predicting properties based on environmental conditions and device settings. Symbolic regression generates mathematical equations $P(\vec{a}, \vec{e}) = S_F$, where $\vec{a}$ represents actuator settings, $\vec{e}$ environmental properties and S_F a feedback state. These equations can be solved to determine the actuator settings $\vec{a}$, required to achieve a desired feedback state S_F . Furthermore, user preferences can be incorporated to select the optimal configuration from a set of possible solutions.

In this paper, we present an outcome-oriented smart home control system, on the example of a light control system for a ray-traced simulated smart home (See Figure 1). Users can manually set the time, lamp, brightness, and window blind positions, and the brightness of the chosen configuration is determined by rendering this scene with a ray-tracer. This is compared to the predicted brightness from our learned equations.

Switching to automatic mode, users set a target brightness, and the application calculates ideal settings for that time, adjusting the blinds and lamp accordingly. This recommendation is also rendered for comparison with the target brightness.

2 RELATED WORKS

2.1 Smart Home Interface

The smart home market is dominated by platforms and hubs developed and maintained by Amazon, Apple and Google[4]. Aside from commercial platforms, there exist open source solutions with

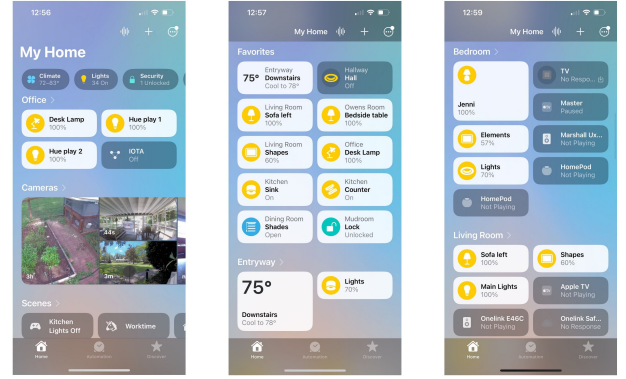


Figure 2: Example images of the Apple Home application[1]. Users can view the status of smart devices, group them by room or function, or change the settings of devices. Already with few devices, it becomes difficult to manage the smart home as users are overwhelmed by options.

consistent user-bases such as OpenHAB[3] or ioBroker[2]. Figure 2 illustrates a typical smart home control application. Smart devices are represented by icons indicating their current state and providing context menus for settings adjustments, such as lamp brightness. Applications from other vendors generally follow this design or offer voice interactions for querying and changing device statuses.

2.2 Outcome-oriented control

King et al. have explored outcome-oriented control using Large-Language-Models (LLMs) to process user requests, like “I’m going to bed” into actions such as “All lights and TV in the living room and entry have been turned off for your bedtime”[6]. While this approach helps users achieve their desired settings, LLMs still suffer from problems such as hallucination[7] and misunderstanding user requests[6]. Our approach fundamentally differs as it involves learning equations that describe the physical environment. These equations are used to determine device settings that achieve a specified target state. In future work, we will compare these approaches in terms of usability and performance metrics.

3 DEMO APPLICATION

In this section, we describe the implementation of our demo application, *SmartBright*. Figure 1 provides an overview of this SwiftUI application. The application is divided into 3 sections:

- **Manual Mode:** the user controls a lamp and the window blinds. The user can decide to turn a device off entirely
- **Room Overview:** the user can observe the perceived brightness P_l of the room. For this demo application, the room is rendered with a ray-tracer and the according P_l is calculated with the following equation:

$$P_l = \log_{10} \left(\sum_{p \in \text{pixels}} 0.2126 * p.r + 0.7152 * p.g + 0.0722 * p.b \right)$$

This equation computes the perceived brightness as the logarithmic sum of the relative luminance of each pixel, based on their red (r), green (g), and blue (b) components[5, 8].

- **Automatic Mode:** the user sets a target brightness, and *SmartBright* solves the equations it previously learned, to find settings that lead to the target brightness of the user. If devices are turned off, *SmartBright* tries to find solutions with the remaining devices, unless the target is not achievable. In that case, additional devices are turned on.
- Additionally, a **time slider** allows to change the current time for the simulation, which changes the skybox for the rendering process and simulates different light conditions.

SmartBright can use one of three methods to learn equations that describe the relationship between the environment, devices, and perceived brightness. We evaluate the performance of these three methods in Section 5.

This feature helps conserve energy by minimizing unnecessary light use. However, if the closest predicted perceived brightness deviates by more than $0.2 P_l$ from the target, the application may override user preferences to meet the primary objective of achieving the desired brightness.

4 METHOD

In this section, we briefly explain our three methods, *Sym*, *SymBS* and *SymUC*. These methods are used to derive equations that describe how time, lamp settings and window blinds affect the perceived brightness P_l in a room. By setting the current time and the target P_l , we can solve these equations to find the necessary smart device settings.

4.1 Sym

Sym uses the complete data set to learn a single equation. The main advantage is that it utilizes 100% of the data for training, ensuring comprehensive data usage.

4.2 SymBS

For data sets with behaviour that varies based on external factors like time, *SymBS* splits the data into intervals of various lengths with shared behaviour. Starting with the index of the lowest unassigned interval, *SymBS* performs a binary search to find the upper boundary of a coherent region. After learning an equation for a region, which consists of at least one interval, the loss is checked against a threshold (0.2 in our case). If the loss is too high, the upper boundary is reduced; if the loss is low enough, the boundary is increased. The advantage of *SymBS* is that equations cover smaller regions. However, data at interval boundaries may behave differently from the rest of the interval, potentially affecting accuracy.

4.3 SymUC

SymUC is similar to *SymBS*, but allows non-contiguous regions to be part of the same region, learning equations for each. *SymUC* performed best in our evaluation, effectively identifying and learning equations for specific times like sunset and sunrise. However, some regions had short durations and insufficient training data. Increasing the sampling rate in these regions could improve training data quality.

5 EVALUATION

In this section, we evaluate *SmartBright*'s ability to predict smart device settings that fulfil user goals. We generated a data set of 10,000 measurements from random smart home configurations in a ray-traced room. In Figure 3 we show the average perceived brightness

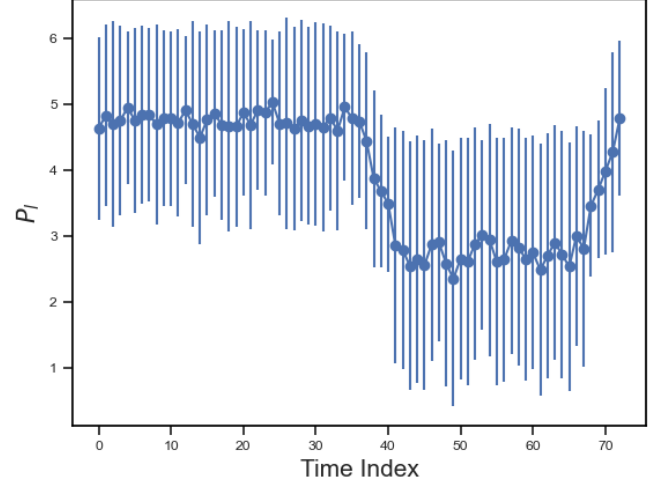


Figure 3: We collected 10,000 measurements of random configurations in a ray-traced room, evenly distributed throughout 72 intervals during a day. Time index 0 is at dawn. We plot for each interval the average perceived brightness P_l , as well as its standard deviation. We clearly see the distinction between day, sunset, night, and sunrise. The standard deviation is noticeably smaller during the day, than during the night.

P_l and its standard deviation σ throughout the day. The average P_l is higher during the day than during the night, as expected. The sun's brightness during the day causes a higher P_l with a lower σ , while at night the lamp significantly impacts P_l , leading to a greater σ .

5.1 Constant Brightness throughout the day

In the first experiment, we set a target brightness P_l of 4.5 in the automatic mode of *SmartBright*. Each method, *Sym*, *SymBS* and *SymUC* was tasked with finding settings to achieve this target. The results are seen in Figure 4. In Table 1 we show the average,

	Average $ P_l $ Error	Min $ P_l $ Error	Max $ P_l $ Error
Sym	0.2686	0.0975	0.4434
SymBS	0.2619	0.035	0.5217
SymUC	0.1163	0.0026	0.4121

Table 1: Average, minimum and maximum absolute distance from target P_l of 4.5 for *Sym*, *SymBS* and *SymUC*. *SymUC* performs best on average, but all methods show similar maximum errors during sunrise/sunset, indicating the need for more training data during these times.

minimum and maximum absolute distance from the chosen target P_l of 4.5 for each methods. *Sym* and *SymBS* perform similarly, whereas *SymUC* performs better overall.

5.2 Constant Brightness without the lamp, if possible

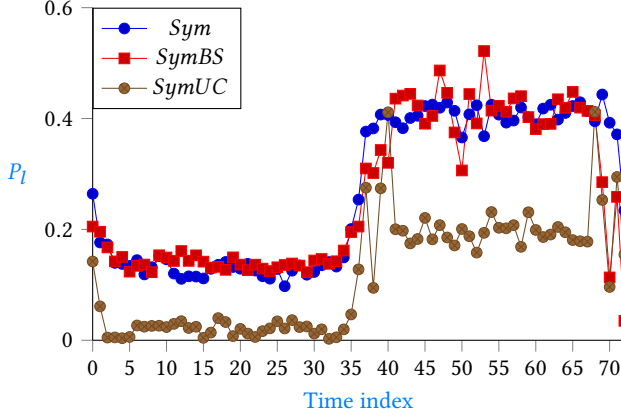


Figure 4: This figure demonstrates that SmartBright can compute nearly optimal settings for window blinds and lamps in a room throughout the day to maintain a desired brightness level. This control mechanism relies on learning a physical model of light scattering. The y-axis represents the error in terms of perceived brightness (P_l), which is calculated as the logarithmic sum of the luminance values of an image. The x-axis represents the time index, ranging from 06:00 AM at index 0 to 05:40 AM at index 72. Each index spans 20 minutes and determines the skybox that is used during ray-tracing. The three curves depict different methods used to learn the physics-based model of the ray-traced environment: *Sym* learns the system holistically, *SymBS* learns coherent regions, and *SymUC* learns non-contiguous regions with similar behavior.

	Average $ P_l $ Error	Min $ P_l $ Error	Max $ P_l $ Error
Sym	0.2775	0.0005	2.2636
SymBS	0.6200	0.0391	3.6545
SymUC	0.1729	0.0139	0.7298

Table 2: Average, minimum and maximum absolute distance from target P_l of 4.5 with lamp usage restricted. *SymUC* shows the lowest average and maximum absolute distances, indicating better performance with restrictions.

We then evaluated user preferences by restricting *SmartBright* to use the lamp only if necessary. If the predicted brightness is more than $0.2 P_l$ away from the target, the lamp is allowed; otherwise, brightness is controlled through the window blind. In Figure 5 and Table 2 we present the results of this evaluation. *SymUC* performs best, finding settings close to the target brightness throughout the

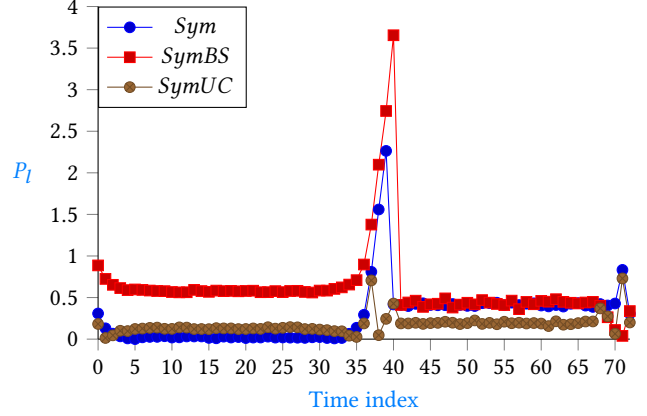


Figure 5: Figure 2 illustrates that SmartBright can incorporate user preferences, such as *Do not use the lamp if possible*, while still maintaining the desired brightness in the room. The three curves again compare the performance of *Sym*, *SymBS*, and *SymUC* for this task. During the day, all three methods use only window blinds to manage the surplus of sunlight. At night, all three methods override the user preference and turn on the lamp. The performance of *Sym* and *SymBS* degrades significantly at the boundary between regions, as they overestimate the available sunlight, resulting in very dark scenes. *SymUC* performs best, as it separates these boundaries into distinct regions and learns individual equations for them.

day. *Sym* and *SymBS* show higher maximum errors, particularly during sunset, due to overestimating the brightness and underestimating the required lamp light.

6 CONCLUSION

Context-aware and outcome-oriented smart home control is crucial for smart home users. We introduced *SmartBright*, a smart home control application that enables an “automatic mode” for lighting in a ray-traced smart home simulation. Users can set a target brightness, and impose restrictions such as “do not use the lamp if possible”, and *SmartBright* finds the necessary settings to achieve the target state. We evaluated three methods, *Sym*, *SymBS* and *SymUC*, which use symbolic regression to enable this application. In future work, we aim to develop a real-world test setup and optimize the hyperparameters of these methods to improve performance further.

REFERENCES

- [1] [n. d.]. Apple’s new Home app in iOS 16 is better but still half-baked. <https://www.theverge.com/23279283/apple-homekit-home-app-ios16-preview>. Accessed: 2024-05-21.
- [2] [n. d.]. ioBroker Automate your life. <https://www.iobroker.net>. Accessed: 2024-05-21.
- [3] [n. d.]. OpenHAB empowering the smart home. <https://www.openhab.org>. Accessed: 2024-05-21.
- [4] [n. d.]. Smart Homes the keys to unlocking smart home ecosystems and catalyzing adoption. <https://harborresearch.com/markets/smart-homes-consumers/>. Accessed: 2024-05-21.

- [5] ITU. 2015. *BT.709 : Parameter values for the HDTV standards for production and international programme exchange*. Standard BT.709-6. International Telecommunication Organization. <https://www.itu.int/rec/R-REC-BT.709-6-201506-1/en>
- [6] Evan King, Haoxiang Yu, Sangsu Lee, and Christine Julien. 2024. Sasha: Creative Goal-Oriented Reasoning in Smart Homes with Large Language Models. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8 (03 2024), 1–38. <https://doi.org/10.1145/3643505>
- [7] Jia-Yu Yao, Kun-Peng Ning, Zhenhui Liu, Munan Ning, and Li Yuan. 2024. LLM Lies: Hallucinations are not Bugs, but Features as Adversarial Examples. <https://openreview.net/forum?id=Rh1aThKliu>
- [8] Peter Zdankin, Viktor Matkovic, Gregor Schiele, and Torben Weis. 2024. Towards Predicting the Perceived Brightness in a Smart Home Through Symbolic Regression. *2024 IEEE International Conference on Pervasive Computing and Communications (PerCom WiP)* X, 22 (2024), X. Preprint:https://anonymous.4open.science/r/ml-1337/PerCom_WiP_2024.pdf